University of Calgary
Faculty of Science
Final Exam
PHYS 365

VERSION 01

December 11, 2025, 5:00–7:00 PM Time: 120 minutes

This is a closed-book exam containing a total of 25 equally weighted questions.

ONLY THE FOLLOWING ITEMS ARE ALLOWED ON YOUR DESK DURING THE EXAM:

1. THIS MULTIPLE-CHOICE QUESTION BOOKLET, which includes the exam questions, the formula sheet, and 3 blank pages. This booklet WILL **NOT** BE HANDED IN.

2. A BUBBLE SHEET used to answer multiple-choice questions. This will be handed in and it WILL BE GRADED.

3. A CALCULATOR. Only a Schulich-approved scientific calculator may be used during the exam.

4. A PEN OR PENCIL, used to black out your answers on the bubble sheet.

5. YOUR STUDENT ID CARD, used to check your identity during exam sign-in.

Students remaining after **105 minutes** must remain seated for the duration of the exam until permitted to leave by the exam supervisor.

1. A source is emitting acoustic waves in a particular medium. Which of the following statements is FALSE?

   A) The speed of the waves is determined by the elastic and inertial properties of the medium.
   B) The speed of the waves is determined by the frequency of the source.
   C) The velocity of the particles of the medium changes with time as the wave passes by.
   D) The wavenumber depends on the medium while the angular frequency does not.

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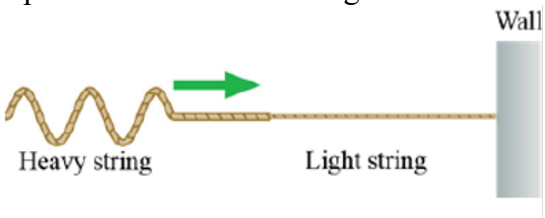
2. Which of the following traveling waves have the same speed?

   I. $D(x,t) = 1.2\ \cos(3x - 2t)$
   II. $D(y,t) = 3.2\ \sin\left(6y + 4t + \frac{\pi}{4}\right)$
   III. $D(y,t) = 3.2\ \sin\left(2y - 4t + \frac{\pi}{4}\right)$
   IV. $D(x,t) = 3.2\ \sin\left(2x + 3t + \frac{\pi}{2}\right)$

   A) I and II
   B) II and III
   C) I and IV
   D) III and IV

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3. Two strings, one heavy and one light, are joined together and the light string is attached to the wall. A travelling wave is sent along the heavy string as shown. The linear mass density of the heavy string is twice as large as the linear mass density of the light string. Compare the mechanical impedances of the two strings.



   A) $Z_{heavy} = \frac{Z_{light}}{2}$
   B) $Z_{heavy} = \frac{Z_{light}}{\sqrt{2}}$
   C) $Z_{heavy} = \sqrt{2}Z_{light}$
   D) $Z_{heavy} = 2Z_{light}$

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4. A grandfather clock uses a pendulum with a period on Earth $T_{Earth} = 2.00$ s. If this clock were placed on Mars, where the gravitational acceleration is $g_{Mars} = 3.728\ \mathrm{m/s^2}$, by how much would the length of the pendulum need to shorten to preserve its period?
The gravitational acceleration of Earth is $g_{Earth} = 9.807\ \mathrm{m/s^2}$

   A) $\Delta L = 0.994$ m

   B) $\Delta L = 0.378$ m

   C) $\Delta L = 0.616$ m

   D) $\Delta L = 0.497$ m

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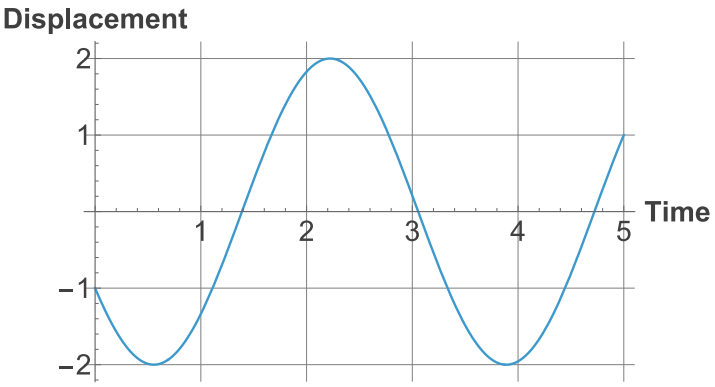
5. A cyclist has two slightly different bells mounted to the handlebars of their bike. They ring both bells at the same time as they travel towards a stationary microphone at 8 m/s. The cyclist hears a tone with a frequency of 257 Hz and 10 beats/s. A stationary microphone is positioned ahead of the cyclist. What frequency does the mic register? Use 340 m/s for speed of sound.

   A) The microphone will register a frequency of 263 Hz, beating with a frequency of 10.24 Hz
   B) The microphone will register a frequency of 251 Hz, beating with a frequency of 9.77 Hz
   C) The microphone will register a frequency of 263 Hz, beating with a frequency of 10 Hz
   D) The microphone will register a frequency of 251 Hz, beating with a frequency of 10 Hz

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6. What is the phase angle of a Simple Harmonic Motion graphically depicted by the image?

   A) $\phi = \frac{2\pi}{3}$
   B) $\phi = -\frac{\pi}{3}$
   C) $\phi = \frac{\pi}{4}$
   D) $\phi = 0$



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7. What are the amplitude and phase angle of a superposition of the following 3 SHMs?

$$y_1(t) = 2\cos(\omega t + \pi/2)$$
$$y_2(t) = 2\cos(\omega t)$$
$$y_3(t) = 3\cos(\omega t + \pi/4)$$

A) $A = 3$ and $\phi = 0$
B) $A = 7$ and $\phi = 3\pi/4$
C) $A = 2.83$ and $\phi = \pi$
D) $A = 5.83$ and $\phi = \pi/4$

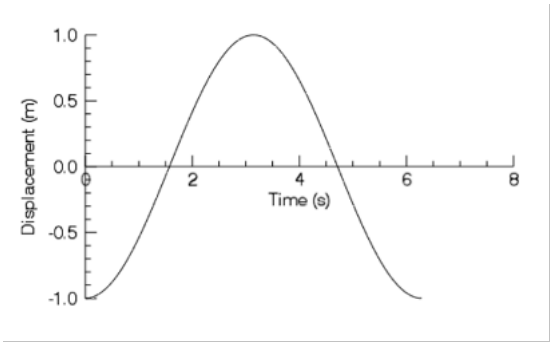
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8. A nylon guitar string is fixed between two lab posts 2.00 m apart. The string has a linear mass density of $\mu = 7.20$ g/m and is placed under a tension of 160 N. The string is placed next to a tube, open at both ends, of length L. The string is plucked and resonate at the fundamental frequency while the tube resonates at the $n = 3$ mode. The speed of sound is 343 m/s. What is the length of the tube?

A) 4.63 m
B) 9.21 m
C) 13.8 m
D) 37.3 m

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9. A standing wave with a node at $x = 0$ is shown. The figure illustrates the displacement as a function of time at the first antinode $\left(x = \frac{\lambda}{4}\right)$. Which of the following equations correctly describes this standing wave?



A) $F(x,t) = 1.0\cos(kx)\cos(\omega t)$
B) $F(x,t) = -1.0\cos(kx)\sin(\omega t)$
C) $F(x,t) = 1.0\sin(kx)\cos(\omega t)$
D) $F(x,t) = -1.0\sin(kx)\cos(\omega t)$

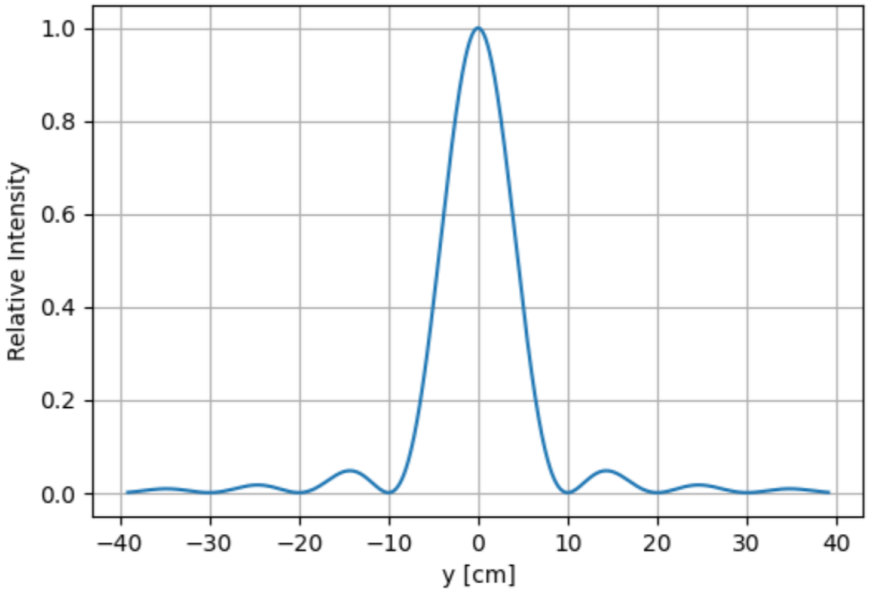
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10. A travelling wave is described by the equation $F(x,t) = 60\cos\big(3\pi(6t+x)\big)$. What are the frequency f and the wavelength λ of the wave?

    A) 10 Hz and 2 m
    B) 2 Hz and10 m
    C) 0.67 Hz and 9 m
    D) 9 Hz and 0.67 m

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11. Light of 450 nm wavelength is shone through a slit to a screen 3.2 m beyond the slit. Based on the diffraction pattern in the graph below, what is the width of the slit?



    A) 14.4 µm
    B) 20 cm
    C) 7.2 µm
    D) 4.5 µm

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12. A string of length 5 m and mass 90 g is under a tension of 100 N. A wave travels along the string and is modeled by the function $y(x,t) = 0.01\sin(15.7x - 1170.12t)$.What is the power transmitted over one wavelength?

    A) 7.3 W
    B) 91.8 W
    C) 9.5 kW
    D) 0.12 kW

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13. An acoustical foam has a density of $29\ \mathrm{kg/m^3}$. What bulk modulus should it have in order to reflect no sound incident normally on its surface? Assume the following properties for air: $B_{air} = 1.42 \times 10^5$ Pa and $\rho_{air} = 1.2\ \mathrm{kg/m^3}$.

    A) 5.88×10^3 Pa
    B) 1.42×10^5 Pa
    C) 1.20×10^3 Pa
    D) 3.00×10^4 Pa

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14. A soap bubble is floating in air. Soap bubbles are a thin film of soap ($n = 1.35$) surrounding air ($n = 1.0$). When light hits the surface of a soap bubble with normal incidence, what is the phase difference **introduced by reflection**?

    A) 0
    B) π
    C) 2π
    D) $\pi/2$

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15. You are performing a double slit-experiment using a laser ($\lambda = 400$ nm) and two slits (each 2.4 μm wide). You do not know the distance between the slits but notice that the $m = 4$ bright fringe is missing, and coincides with the first diffraction minimum. What is the distance between the slits?

    A) 1.6 μm
    B) 7.2 μm
    C) 9.6 μm
    D) 19.2 nm

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16. Consider an electron ($m_e = 9.11 \times 10^{-31}$ kg) that has the same de Broglie wavelength as a neutron ($m_n = 1.67 \times 10^{-27}$ kg). Which of the following statements is true?

    A) The neutron would have greater kinetic energy, speed, and momentum than the electron
    B) The neutron and electron would have the same momentum, but the electron would have a greater speed and kinetic energy.
    C) The neutron and electron would have the same momentum but the neutron would have a greater speed and kinetic energy
    D) The electron would have a greater speed, kinetic energy, and momentum than the neutron.

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17. A hydrogen atom is in its second excited state. Which energy level should it transition to in order to emit light of the longest possible wavelength?

    A) $n = 0$
    B) $n = 1$
    C) $n = 2$
    D) $n = 3$

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18. An electron is confined in a one-dimensional infinite potential well. If the energy of the electron at the $n = 8$ energy level is measured to be 14.10 eV, what is the width L of the well?

    A) 0.42 nm
    B) 1.31 nm
    C) 2.62 nm
    D) 10.48 nm

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19. In microscopy, the Heisenberg uncertainty principle limits how precisely atomic positions can be measured. When a sample is observed, photons scatter off the atoms, perturbing their motion. A photon with wavelength $\lambda = 440$ nm scatters off an atom in a crystalline lattice. Using the uncertainty principle, determine the minimum uncertainty in the atom's position.

    A) 35 nm
    B) 75 nm
    C) 110 nm
    D) 220 nm

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20. You have two sources of radiation: Source A has a wavelength 520 nm, and Source B has a wavelength of 300 nm. Source A emits light that is twice the intensity of Source B. You direct both sources at a sheet Aluminum ($\Phi = 4.08$ eV). You measure the stopping voltage and photocurrent in both cases. Which of the following statements is true?

    A) The stopping voltage will be lower for Source A, but the photocurrent will be higher
    B) A photocurrent and stopping voltage will only be measurable for source B
    C) The stopping voltage will be higher for source A
    D) The photocurrent will be higher for source A

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21. An electron confined to a 1-dimensional infinite potential well is in the $n = 3$ energy state. Where is it most likely to be found?

A) At the center of the box
B) One-quarter and three-quarters of the way from the ends
C) One-sixth, one-half, and five-sixths of the way from the ends
D) Equally likely to be found anywhere in the box

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22. An electron is confined in a one-dimensional infinite potential well of width $L = 200$ pm. The electron is in the third excited state. What is the probability of finding the electron in the leftmost eighth of the well.

A) 0.091
B) 0.91
C) 0.125
D) 0.25

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23. 6 MeV X-rays are scattered off a carbon-12 target. What is the shifted wavelength λ' that will be measured at a scattering angle of 30º?

A) 5.3×10^{-13} m
B) 4.0×10^{-13} m
C) 8.6×10^{-13} m
D) 2.1×10^{-13} m

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24. Rank the Compton shift that would be measured at the following angles from smallest to largest: 0º, 45º, 150º, and 210º.

A) 45º < 0º < 150º < 210º
B) 150º < 210º < 0º < 45º
C) 150º = 210º < 0º < 45º
D) 0º < 45º < 150º = 210º

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25. What potential difference would accelerate an electron such that it has a wavelength of 5.48×10^{-11} m?

A) 1.2 kV
B) 75 kV
C) 50 kV
D) 0.5 kV

PHYS 365 Final Exam Formula Sheet

OSCILLATIONS

$x(t) = x_m \cos(\omega t + \phi)$	$\omega = 2\pi f = \frac{2\pi}{T}$	$\omega = \sqrt{\frac{k}{m}}$	$\omega = \sqrt{\frac{g}{\ell}}$	$T = 2\pi\sqrt{\frac{\ell}{g}}$

WAVES

$\frac{\partial^2 y}{\partial^2 t} = v^2 \frac{\partial^2 y}{\partial x^2}$	$y(x,t) = f(x \pm vt)$	$y(x,t) = y_m \cos(kx \pm \omega t + \phi)$	
$v = \frac{\omega}{k} = f\lambda$	$k = \frac{2\pi}{\lambda}$	$v = \frac{c}{n}$	$\lambda = \frac{\lambda_0}{n}$
$v = \sqrt{\frac{\tau}{\mu}}$	$v = \sqrt{\frac{\gamma RT}{M}}$	$v = \sqrt{\frac{Y}{\rho}}$	$v = \sqrt{\frac{B}{\rho}}$
$Z = \sqrt{\mu\,\tau} = \frac{\tau}{v} = \mu v$	$Z = \sqrt{\rho B} = \rho v$	$P_{avg} = \frac{1}{2}\mu v \omega^2 y_m^2$	$\Delta p_m = (v\rho\omega)s_m$
$\Delta p = \Delta p_m \sin(kx - \omega t)$	$I = \frac{1}{2}\rho v \omega^2 s_m^2$	$I = \frac{P}{4\pi R^2}$	$I = \frac{P}{A}$
$\beta = (10\ dB)\log_{10}\left(\frac{I}{I_0}\right)$	$\Delta\beta = (10\ dB)\log_{10}\left(\frac{I_2}{I_1}\right)$	$I(r_2) = \left(\frac{r_1}{r_2}\right)^{n-1} I(r_1)e^{-\alpha\Delta r}$	
$R = \frac{Z_1 - Z_2}{Z_1 + Z_2}$	$T = \frac{2Z_1}{Z_1 + Z_2}$	$B = -\frac{\Delta p}{\left(\frac{\Delta V}{V}\right)}$	$Y = \frac{\left(\frac{F}{A}\right)}{\left(\frac{\Delta L}{L}\right)}$
$R_e = R^2$	$T_e = \frac{Z_2}{Z_1}T^2$	$\Delta p = -B\frac{\partial s}{\partial x}$	

WAVE INTERFERENCE

$y'(x,t) = \left[2y_m \cos\left(\frac{1}{2}\phi\right)\right]\sin\left(kx - \omega t + \frac{1}{2}\phi\right)$		$y'(x,t) = [2y_m \sin(kx)]\cos(\omega t)$	
$\Delta\phi = \frac{\Delta L}{\lambda}2\pi$	$f_{beat} = f_1 - f_2$	$f' = f\frac{v \pm v_D}{v \pm v_S}$	$f' = \sqrt{\frac{c \pm v}{c \mp v}}f$
Constructive interference $\frac{\Delta L}{\lambda} = 0,\ 1,\ 2, \ldots$		Destructive interference $\frac{\Delta L}{\lambda} = 0.5,\ 1.5,\ 2.5, \ldots$	
Taut string or pipe open at both ends $f_n = \frac{v}{\lambda_n} = \frac{nv}{2L}$ for $n = 1,\ 2,\ 3, \ldots$		Pipe one end closed, one end open $f_n = \frac{v}{\lambda_n} = \frac{nv}{4L}$ for $n = 1,\ 3,\ 5, \ldots$	
$d\sin(\theta) = m\lambda$ for $m = 0, 1, 2, \ldots$ (maxima-bright fringes)			
$d\sin(\theta) = \left(m + \frac{1}{2}\right)\lambda$ for $m = 0, 1, 2, \ldots$ (minima-dark fringes)			

$2t = m\frac{\lambda_0}{n}, \quad m = 1, 2, 3 \ldots$		$2t = \left(m - \frac{1}{2}\right)\frac{\lambda_0}{n}, \quad m = 1, 2, 3 \ldots$	
Reflection	Reflection phase shift	$a\sin(\theta) = m\lambda$ for $m = 1, 2, 3, \ldots$ (minima)	
Off lower index	0		
Off higher index	0.5 wavelength		
$I = 4I_0\cos^2\left(\frac{1}{2}\phi\right)$	$\phi = \frac{2\pi d}{\lambda}\sin(\theta)$	$\theta_R = \sin^{-1}\left(\frac{1.22\lambda}{d}\right)$	
MODERN PHYSICS			
$F_e = \frac{1}{4\pi\epsilon_0}\frac{e^2}{r^2}$	$U = \frac{1}{4\pi\epsilon_0}\frac{e^2}{r}$	$U = qV$	$K = \frac{1}{2}mv^2$
$E = hf$	$p = \frac{h}{\lambda}$	$hf = K_{max} + \Phi$	$a_c = \frac{v^2}{r}$
$\Delta\lambda = \frac{h}{mc}(1 - \cos\phi)$	$\frac{1}{\lambda} = R\left(\frac{1}{n_{low}^2} - \frac{1}{n_{high}^2}\right)$	$K_{max} = eV_{stop}$	$R = \frac{P}{hf}$
$\Delta x\Delta p_x \geq \frac{h}{4\pi}$	$a_0 = \frac{\epsilon_0 h^2}{\pi m_e e^2}$	$E_n = -\frac{m_e e^4}{8\epsilon_0^2 h^2 n^2}$	$E_n = \frac{-13.6\text{ eV}}{n^2}$
$E_n = \frac{h^2}{8m_e L^2}n^2$	$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi}{L}x\right)$	$P(x_1, x_2) = \int_{x_1}^{x_2}\lvert\psi(x)\rvert^2 dx$	
$P(r) = \frac{4}{a_0^3}r^2 e^{\frac{-2r}{a_0}}$	$\psi(r) = \frac{1}{\sqrt{\pi}a_0^{3/2}}e^{\frac{-r}{a_0}}$	$T \approx e^{-2bL}$ $b = \sqrt{8\pi^2 m_e(U_b - E)/h^2}$	
$\frac{d^2\psi}{dx^2} + k^2\psi = 0, \quad \text{with} \quad k = \frac{2\pi\sqrt{2m_e(E-U)}}{h}$		$\frac{d^2\psi}{dx^2} + \frac{8\pi^2 m_e}{h^2}\big(E - U(x)\big)\psi = 0$	
CONSTANTS			
$c = 3.00 \times 10^8\ \frac{\text{m}}{\text{s}}$	$R = 8.314\ \frac{\text{J}}{\text{mol}\cdot\text{K}}$	$M_{air} = 28.8\ \frac{\text{g}}{\text{mol}}$	$g = 9.81\ \frac{\text{m}}{\text{s}^2}$
$v_{sound\ at\ 20°\text{C}} = 343\ \frac{\text{m}}{\text{s}}$	$T_K = T_{°\text{C}} + 273.15$	$\gamma_{monoatomic} = \frac{5}{3}$	
$I_0 = 1 \times 10^{-12}\ \frac{\text{W}}{\text{m}^2}$	$1\text{ atm} = 1.01 \times 10^5\text{ Pa}$	$\gamma_{diaatomic} = \frac{7}{5}$	
$m_e = 9.109 \cdot 10^{-31}\text{ kg}$		$m_p = 1.67 \cdot 10^{-27}\text{ kg}$	
$a_0 = 0.5292 \cdot 10^{-10}\text{ m}$		$1\ u = 1.66054 \cdot 10^{-27}\text{ kg}$	

$h = 6.63 \cdot 10^{-34}\ \mathrm{J \cdot s} = 4.136 \cdot 10^{-15}\ \mathrm{eV \cdot s}$	$\epsilon_0 = 8.854 \cdot 10^{-12}\ \frac{\mathrm{C^2}}{\mathrm{N \cdot m^2}}$
$\sigma = 5.67 \cdot 10^{-8}\ \frac{\mathrm{W}}{\mathrm{m^2 \cdot K^2}}$	$e = 1.602 \cdot 10^{-19}\ \mathrm{C}$
$k_B = 1.381 \cdot 10^{-23}\ \frac{\mathrm{J}}{\mathrm{K}} = 8.617 \cdot 10^{-5}\ \frac{\mathrm{eV}}{\mathrm{K}}$	Rydberg constant $R = 1.09737 \cdot 10^{7}\ \mathrm{m^{-1}}$
MATHEMATICAL FORMULAE	
$\cos(\alpha) \pm \cos(\beta) = 2\cos\left(\frac{1}{2}(\alpha \pm \beta)\right)\cos\left(\frac{1}{2}(\alpha \mp \beta)\right)$	
$\sin(\alpha) \pm \sin(\beta) = 2\sin\left(\frac{\alpha \pm \beta}{2}\right)\cos\left(\frac{\alpha \mp \beta}{2}\right)$	
$\cos(\alpha \pm \beta) = \cos(\alpha)\cos(\beta) \mp \sin(\alpha)\sin(\beta)$	$\tilde{A} = Ae^{j\theta} = A(\cos(\theta) + j\sin(\theta))$
$\sin(\alpha \pm \beta) = \sin(\alpha)\cos(\beta) \pm \cos(\alpha)\sin(\beta)$	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
$\sin(-\alpha) = -\sin(\alpha)$	$\cos\left(\alpha \pm \frac{\pi}{2}\right) = \mp\sin(\alpha)$
$\cos(-\alpha) = \cos(\alpha)$	$\sin\left(\alpha \pm \frac{\pi}{2}\right) = \pm\cos(\alpha)$
$\int \cos(ax) = \frac{\sin(ax)}{a} + c$	$\int \sin(ax) = -\frac{\cos(ax)}{a} + c$
$\int \sin^2(ax)\,dx = \frac{x}{2} - \frac{\sin(2ax)}{4a} + c$	$\int \cos^2(ax)\,dx = \frac{x}{2} + \frac{\sin(2ax)}{4a} + c$
$\sin^2\theta = \frac{1}{2}(1 - \cos(2\theta))$	$\cos^2\theta = \frac{1}{2}(1 + \cos(2\theta))$